

A proof of Conjecture 00000004930

Two MDPs with equal discounted values for every discount factor but different average rewards

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1 The conjecture

Conjecture 00000004930 reads (English version; the Chinese version says the same, with “all discount-factor values identical”):

Definition: Discounted value and average reward are two layers of optimality measures.

Conjecture: There exist two MDPs with identical values for all discount factors but different average rewards, and the separation is realized by an explicit construction concentrating the discount weights on transients.

It is an existence statement. We give two explicit Markov decision processes with countable state spaces whose optimal discounted values agree for every discount factor, while under every policy the long-run average reward is 1 in the first and 0 in the second. The second MDP collects all of its reward on transient states. Everything is proved in Lean 4 with Mathlib.

2 Definitions

Definition 1 (MDP, policies). An MDP M (Lean: `MDP`) consists of a state type S , an action type $\mathcal{A}$, an initial state s_0 , transition laws $P(s, a) \in \text{PMF}(S)$ and rewards $r(s, a) \in \mathbb{R}_{\geq 0}$. A history at time t is $((s_0, a_0), \dots, (s_{t-1}, a_{t-1}); s_t)$ (Lean: `Hist`). A policy is history-dependent and randomized: $\pi(h) \in \text{PMF}(\mathcal{A})$ (Lean: `Policy`). The law of the history at time t is defined recursively by s_0 at $t = 0$ and, from a history h with current state s , by drawing $a \sim \pi(h)$, $s' \sim P(s, a)$ and appending (s, a) (Lean: `law`).

Definition 2 (Values and average reward). The expected reward at time t is $\mathbb{E}^\pi[r(s_t, a_t)]$ (Lean: `expReward`). For a discount factor β , the discounted value of π is $V_\beta(\pi) = \sum_{t \geq 0} \beta^t \mathbb{E}^\pi[r(s_t, a_t)]$ (Lean: `discValue`), and the (optimal) value of M is $V_\beta^* = \sup_\pi V_\beta(\pi)$ (Lean: `optValue`). The T -step average is $\frac{1}{T} \sum_{t < T} \mathbb{E}^\pi[r(s_t, a_t)]$ (Lean: `avgReward`); the average reward of π is its limit (or $\liminf$, or $\limsup$) as $T \rightarrow \infty$, and the optimal average reward is the supremum over policies. Lean computes the T -step average from the real numbers (`expReward M π t`).`toReal`; since `toReal` sends ∞ to 0, this agrees with the usual definition whenever the expected rewards are finite, e.g. for bounded rewards. Both witnesses have rewards in $\{0, 1\}$, so their expected rewards are at most 1 (`expReward_le_one`) and the formal and usual averages coincide.

Values are taken in $[0, \infty]$. The Lean statements let β range over all of $[0, \infty]$ with the arithmetic of $[0, \infty]$, in which $1 - \beta$ is truncated subtraction; so the Lean value $(1 - \beta)^{-1}$ means $1/(1 - \beta)$ for a discount factor $\beta \in [0, 1)$ and ∞ for $\beta \geq 1$ (where the series $\sum_t \beta^t$ indeed diverges). Below we write $(1 - \beta)^{-1}$ in this sense; for every discount factor $\beta \in [0, 1)$ it is the ordinary number $1/(1 - \beta)$.

3 The two MDPs

MDP A (Lean: A): one state, one action, reward 1.

MDP B (Lean: B): states $start$, $count(k)$ for $k \in \mathbb{N}$, and $sink$; actions $n \in \mathbb{N}$; deterministic transitions (Lean: `next`)

$$start \xrightarrow{n} count(n), \quad count(k+1) \rightarrow count(k), \quad count(0) \rightarrow sink, \quad sink \rightarrow sink,$$

where the action matters only at $start$; reward 1 at every state except $sink$, where it is 0. Playing n first yields reward 1 at times $0, 1, \dots, n+1$ and 0 afterwards. All rewarded states are transient: each is visited at most once, and the discount weights β^t that make up V_β are carried by this transient prefix.

4 Results

Lemma 3 (`optValue_A`, `avgReward_A`). *In A every policy has expected reward 1 at every time. Hence $V_\beta^*(A) = (1 - \beta)^{-1}$ for every β , and every policy has average reward 1.*

Lemma 4 (`optValue_B`). $V_\beta^*(B) = (1 - \beta)^{-1}$ for every β .

Proof. Rewards are at most 1, so the expected reward is at most 1 (`expReward_le_one`) and $V_\beta(\pi) \leq \sum_t \beta^t$. Conversely the deterministic policy that always plays n (`detPol`) has expected reward 1 at times $t \leq n+1$ (`expReward_detPol`), so $V_\beta^*(B) \geq \sum_{t \leq n} \beta^t$ for every n , whose supremum is $\sum_t \beta^t = (1 - \beta)^{-1}$. $\square$

Lemma 5 (`law_state`). *Let π be any policy for B and $q = \pi(start)$ the law of the first action. For every $t \geq 1$ the law of the state s_t is the image of q under $n \mapsto st(n, t)$, where $st(n, \cdot)$ is the state sequence obtained by playing n .*

Proof. Induction on t . For $t = 1$ this is the first transition. If the claim holds at $t \geq 1$, then every history in the support of the time- t law has state $\neq start$; at such states the transition does not depend on the action, so the randomized, history-dependent choice of a_t is irrelevant and the state law is pushed forward by the fixed step map. $\square$

Lemma 6 (`expReward_B_le`, `expReward_B_tendsto`, `avgReward_B`). *Under every policy for B: $\mathbb{E}^\pi[r_{t+1}] \leq \sum_k q(k+t)$, so the expected reward tends to 0, and by Cesaro's theorem the average reward is 0.*

Theorem 7 (`conjecture4930`). *For every β : $V_\beta^*(A) = V_\beta^*(B) = (1 - \beta)^{-1}$. Under every policy the average reward of A tends to 1 and that of B tends to 0; in B the expected reward at time t tends to 0 under every policy. Hence A and B have identical values for all discount factors and different (optimal) average rewards, $1 \neq 0$.*

5 Remarks on the reading

- *Values* are optimal values at the initial state, and *average rewards* are long-run average rewards; since the limit exists for every policy, the $\lim$, $\liminf$ and $\limsup$ versions and their suprema over policies all give 1 for A and 0 for B.

- The separation needs a countably infinite state space. For finite MDPs it is impossible, because $g^* = \lim_{\beta \uparrow 1} (1 - \beta)V_\beta^*$ (Blackwell); for a single fixed policy it is impossible as well, because $\beta \mapsto V_\beta$ is the generating function of the expected reward sequence, which it therefore determines. The conjecture does not restrict to finite MDPs, and its phrase “concentrating the discount weights on transients” describes exactly this construction: all reward of B lies on transient states, which the discounted criterion sees and the average criterion ignores.
- Rewards are taken nonnegative (both witnesses have rewards in $\{0, 1\}$); policies are fully general (history-dependent and randomized).

6 Lean formalization and axiom audit

The file `lean/Conjecture4930/Basic.lean` (300 lines, Lean 4.33.1, Mathlib v4.33.1) contains the definitions above and the theorems `optValue_A`, `avgReward_A`, `optValue_B`, `law_state`, `expReward_B_le`, `expReward_B_tendsto`, `avgReward_B` and the main theorem `Conjecture4930.conjecture4930`. It builds with `lake build`. `#print axioms conjecture4930` (and `law_state`) reports exactly `[propext, Classical.choice, Quot.sound]`. There is no `sorry`, no `native_decide` and no added axiom.